Aaditya Chopra

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Education

University of Washington — **Seattle, WA, USA**
M.S. in Data Science — Sep 2024 – Mar 2026

Thapar Institute of Engineering and Technology — **Patiala, India**
B.E. in Computer Science — Jul 2017 – Jun 2021

Technical Skills

Languages: Python, C, C++, SQL, TypeScript, R, JavaScript
Frameworks: PyTorch, TensorFlow, Keras, PySpark, Pandas, NumPy, Polars, Docker, AWS
Coursework: Machine Learning, Computer Vision, NLP, Interactive Learning, Statistics, Probability, A/B Testing, Causal Inference, Distributed Computing, Data Structures, Algorithms, Database Management

Experience

Corvic.ai — **Remote, USA**
Machine Learning Engineer Intern — Sep 2025 – Dec 2025

- Architected **LLM-powered AI agent orchestrator** for autonomous multi-step reasoning, implementing tool-calling validation, reflection loops, and chain-of-action generation enabling zero-shot data analysis across enterprise datasets; **outperformed vanilla GPT-4, Reflexion, and LangChain on reasoning and Q&A benchmarks**.
- Engineered multi-modal image embedding pipeline with **CLIP-ViT** and **SigLIP2** through **Temporal workflows**, optimizing batched vector generation that reduced embedding latency by **60%** while processing **10K+** images daily.

Databricks — **Mountain View, CA, USA**
Data Science Intern — Jun 2025 – Sep 2025

- Developed **anomaly detection** model to flag storage, compute, and network abuse in community edition accounts.
- Co-designed a **Neo4j** property graph for Trust & Safety signals, enabled **multi-hop path queries** and **GDS community detection**, cutting investigation time by **13 hrs/week**.
- Built attribution logic using **Node2Vec** on account-level subgraphs to surface abuse cohorts and enrich alerts.

JP Morgan Chase & Co. — **Mumbai, India**
Quantitative Research Associate — Jan 2024 – Aug 2024

- Engineered an anomaly detection system, monitoring **150,000+** time series daily and pinpointing statistical anomalies.
- Built a scalable, fault-tolerant, cross-vendor framework using **ARIMA** for time series forecasting, providing key insights on Average Daily Traded Volume to the firm's **Credit Officers** for trade approvals and liquidity add-on calculations.
- Revamped the Data Quality Program (DQP), **reducing $15M** in market risk capital by streamlining infrastructure.
- Provided mentorship and facilitated seamless onboarding for new team members.

Quantitative Research Analyst — Jul 2021 – Dec 2023

- Leveraged **DBSCAN clustering** for anomaly detection, identifying high-risk trades and improving risk management.
- Applied **Quantile Regression** and other advanced statistical techniques to enhance Value at Risk (VaR) calculations.

Quantitative Research Intern — Jan 2021 – Jun 2021

- Automated time series monitoring using statistical models and Python **saving 80% of manual effort** across the team.

Projects

Neural Encoding of Auditory-Linguistic Features via EEG Modeling | mTRF Modeling — Ongoing

- Developed **EEG modeling** pipeline combining linguistic annotations with auditory features to study neural encoding
- Analyzed spatial EEG activations to extract population-level neural response patterns; **work contributed to a neuroscience conference manuscript**.

MICCAI Unicorn Challenge | *Fine Tuning LLM* — Jun 2025 – Aug 2025

- Fine-tuned **multi-modal LLMs** (MedImageInsight, UltraMed) and **transformers** (LLaMA, RoBERTa), applying cross-modal learning with efficient transformer backbones for **20** pathology and radiology tasks under compute limits.
- Worked as Research Assistant at KurtLab (**UW Medicine**) and achieved **5%** improvement over baseline.

MRI to PET Image Synthesis | *Python, Tensorflow, OpenCV* | Link — Feb 2025

- Developed a **Stochastic Differential Equation** (VESDE) based **Diffusion Model** with **U-Net architecture** to transform MRI scans into PET images, offering a cost-effective alternative to expensive PET scans.
- Achieved a PSNR of 33 dB, **20% better** than conventional methods, exhibiting significant image quality improvements.